A Review of the Legal Framework Governing E-Waste Management, its Impacts and Disposal

Kassim Ghanchi
Roll No. 24102C0015
Department of Computer Engineering, Third Year

***Abstract*—**This paper reviews the legal and regulatory framework governing electrical and electronic waste (e-waste), drawing on Chadha and Chadha's analytical study of e-waste generation, its hazardous composition, and its impacts on human health and the environment. It surveys the principal international instruments — the Basel, Bamako and Rotterdam Conventions, the EU's WEEE and RoHS Directives — alongside the evolution of India's domestic e-waste regime from the Factories Act, 1948 through the E-Waste (Management and Handling) Rules, 2011 and their 2016 and 2018 amendments. The review consolidates the source study's findings on recycling gaps, the role of the informal sector, and cross-border movement of hazardous waste into India, and closes with its recommendations for strengthening Extended Producer Responsibility (EPR)-based enforcement.

Index Terms*—E-waste, Extended Producer Responsibility, Basel Convention, RoHS, hazardous waste, E-Waste (Management) Rules.*

I. Introduction

The rapid expansion of electrical and electronic equipment (EEE) — accelerated further by the Internet of Things and "smart" devices — has a dual effect on the environment: it multiplies the volume of equipment discarded annually, and it drives extraction of raw materials to manufacture replacements [1]. The Solving the E-waste Problem (StEP) Initiative (2014) defines e-waste as "all items and parts of electronic and electric equipment which have been disposed off by its owner as waste without the intention of reuse." Because devices manufactured after the growth of complex electronics can contain over 1,000 distinct substances, several of them toxic, their disposal creates a pollution problem that is comparatively recent in origin [1]. This review consolidates the legal responses — international and Indian — that have emerged to manage this problem, together with the health, environmental and enforcement challenges that remain.

II. E-Waste Generation and Global Trends

According to the Global E-waste Monitor 2020, 53.6 Mt of e-waste were generated worldwide in 2019 — an average of 7.3 kg per capita — and this figure is projected to nearly double, to 74.7 Mt, by 2030 [1]. Asia generated the largest share (24.9 Mt), followed by the Americas (13.1 Mt) and Europe (12 Mt), with Africa and Oceania contributing 2.9 Mt and 0.7 Mt respectively. Shorter device life-cycles and limited repair options are identified as the principal drivers of this growth [1].

India generated 3.2 Mt of e-waste in 2019, ranking third globally after China and the United States [1]. Table I reproduces the year-wise national generation trend reported in the source study.

TABLE I. YEAR-WISE E-WASTE GENERATION IN INDIA (MILLION METRIC TONNES)

Year	Generation (MMT)
2015	1.97
2016	2.22
2017	2.53
2018	2.86
2019	3.23

Source: CSE, 2020, compiled from the Global E-waste Statistics Partnership [1]. Separately, the Central Pollution Control Board estimated national generation for 2019–20 at 10,14,961.2 tonnes across 21 EEE categories, based on sales data from 1,380 producers [1].

III. Health and Environmental Impacts

By composition, roughly 50% of e-waste is iron and steel, 21% plastics, and 13% non-ferrous metals, alongside printed circuit boards, glass and other constituents [1]. Hazardous waste within this stream — including lead, arsenic, mercury and PCBs — produces primary emissions on disposal; incomplete treatment generates secondary emissions such as dioxins and furans; and the chemicals used in recycling itself produce tertiary emissions [1].

Substances such as antimony, beryllium, brominated flame retardants, cadmium, hexavalent chromium and mercury — found in circuit boards, batteries, CRTs and integrated circuits — are persistent and bio-accumulative. Documented health effects include adverse birth outcomes, altered neuro-development, DNA damage, cardiovascular and respiratory harm, immune-system effects, skin disease, hearing loss and cancer [1].

Recycling capacity has not kept pace with generation: only 9.3 Mt of e-waste (17.4% of global generation) was formally collected and recycled in 2019. Europe led at a 42.5% collection rate, followed by Asia (11.7%); Africa recorded the lowest rate, at 0.9% [1]. In India, only about one-quarter of the 3.2 Mt generated in 2019 could be recycled domestically, and the country has historically also received imported hazardous waste — 6.4 Mt imported in 2009 against 5.9 Mt generated domestically — including mercury and electronic waste from the United States, asbestos from Canada, and metal scrap from several European states [1].

IV. International Legal Framework

A cluster of multilateral instruments underpins global e-waste governance. The Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal (1989, in force 1992) has been signed by 187 countries and led, via its 2006 Conference of the Parties, to the Nairobi Declaration on environmentally sound e-waste management [1]. The Bamako Convention (1991/1998) extends similar controls across Africa without the exceptions the Basel Convention allows for radioactive material, while the Rotterdam Convention regulates trade in hazardous chemicals through a Prior Informed Consent procedure — India acceded to it on 24 May 2005 [1].

Within the European Union, the WEEE Directive (2012/19/EU) prohibits the export of hazardous waste from member states to developing countries, while the RoHS Directive (2002/95/EC, in force February 2003; now RoHS 3) restricts hazardous substances in EEE manufacture [1]. The European Commission's 2000 Communication on the Precautionary Principle further shapes how member states assess and manage residual risk [1]. At the institutional level, the US EPA, StEP (formerly UNU-Step) and UNCTAD coordinate cross-border research, awareness and enforcement efforts, though Greenpeace has separately argued for an outright ban on used-electronics exports [1].

National approaches vary widely: most US states use an EPR model in the absence of federal legislation; Canada regulates through 12 provinces and territories; several Latin American states (Brazil, Argentina, Ecuador, Bolivia) have adopted or drafted EPR-based waste laws; and across Asia, China, Japan, South Korea and Israel (from 2014, with a 50%-by-weight recycling mandate from 2021) have each introduced dedicated e-waste legislation, with India identified as the only country with such legislation in South Asia [1].

V. Legal Framework in India

India's e-waste regime has developed incrementally. The Factories Act, 1948 first listed contaminants arising from manufacture or recycling of electronic components; the Environmental Protection Rules, 1986 brought electro-plating units under effluent-disposal norms; and the Hazardous Waste (Management and Handling) Rules, 1989 (amended 2003) applied specific Basel-linked schedule entries to e-waste disposal and recycling [1].

The E-Waste (Management and Handling) Rules, 2011 — superseded in 2016 — introduced the Extended Producer Responsibility principle for the first time, though it excluded informal-sector workers, and made registration of dismantlers and recyclers with State Pollution Control Boards obligatory [1]. The Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016 banned import of e-waste from developed countries except for refurbishment of second-hand goods, which must in turn be re-exported within one to three years [1]. A 2016 Parliamentary Committee report subsequently recommended more robust infrastructure and enforcement mechanisms, and the E-Waste (Management) Amendment Rules, 2018 required e-waste to be channelled specifically to authorised dismantlers and recyclers [1]. The Ministry of Electronics and Information Technology has separately run an E-waste Awareness programme since 2015 under the ‘Swachh Digital Bharat’ banner [1].

This trajectory continues in the E-Waste (Management) Rules, 2022, which consolidate EPR into a portal-based system: manufacturers, producers, refurbishers and recyclers must register and meet phase-wise recycling targets — 60% of generation for FY 2023–24 and 2024–25, rising to 70% and then 80% by FY 2027–28 — enforced through EPR certificates purchased from registered recyclers [2].

VI. Challenges in E-Waste Management

The source study identifies several persistent obstacles. First, the bulk of India's e-waste collection, transport and recycling still runs through an unregulated informal sector — well networked but without safety guidelines — exposing workers and, often, children living near recycling sites to direct hazard [1]. Second, recycling infrastructure remains absent or underdeveloped in many middle- and low-income countries. Third, weak inspection capacity at customs and the Ministry of Environment, Forest and Climate Change makes it difficult to distinguish genuine second-hand goods from disguised e-waste, and India's law is described as silent on penalties even where such fraud is detected [1]. Fourth, most State Pollution Control Board websites do not publish basic information on registered recyclers and collectors, limiting public accountability [1].

VII. Findings and Recommendations

The study's recommendations centre on strengthening enforcement rather than adding new instruments. It proposes that Producer Responsibility Organisations — modelled loosely on Norway's take-back system but adapted to Indian conditions — be made financially, physically and legally responsible for products at end-of-life [1]. It further recommends prohibiting e-waste disposal in municipal landfills; applying Life Cycle Assessment, Material Flow Analysis and Multi-Criteria Analysis to guide policy; introducing deposit-refund schemes at the point of purchase; providing separate e-waste bins with penalties for misuse; and using RFID tagging on circuit boards and refrigerant compressors to track products across their life cycle [1]. On cross-border movement, it recommends that any country responsible for disposal receive a fee of 2–5% of original production cost, paid into a fund that subsidises certified processors by verified throughput [1]. It also calls for formal integration of the informal recycling sector rather than its exclusion, so that safety and fair labour conditions can be achieved without eliminating the livelihoods it supports [1].

VIII. Conclusion

Considered together, the international conventions and India's own rules from 1948 through 2022 show a clear, if gradual,

shift toward Extended Producer Responsibility as the organising principle for e-waste governance. What remains outstanding, on this review's evidence, is enforcement: recycling rates trail generation by a wide margin worldwide, the informal sector continues to absorb a large share of India's e-waste outside any regulatory line of sight, and monitoring infrastructure — both physical and informational — lags behind the rules already on the books. Closing that gap, more than further legislation, is what the source study identifies as the precondition for a genuinely circular approach to electronic waste.

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